

Text Classification in NLP

CMPE346 · Group 2

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Group task

Text Classification

Metric

Accuracy & F1

Student	Task
Atakan Gül	Media Bias Detection
Ece Doğa Gül	Sexism Detection
Talha Demir	Ghost-Job Risk Classification

BiLSTM (baseline)

Bidirectional recurrent network trained from scratch — no pretraining.

BERT `bert-base-uncased` & RoBERTa `roberta-base`

Pretrained transformer encoders fine-tuned end-to-end; RoBERTa adds dynamic masking and removes the NSP objective.

Task

Classify a news sentence as *biased* or *not biased*.

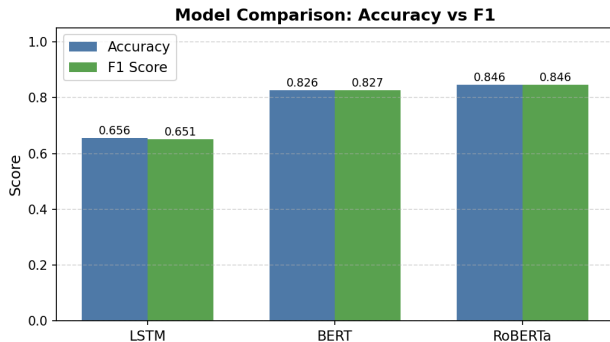
Dataset: BABE

- 4,121 expert-annotated sentences
- Diverse news outlets
- 3,121 train / 1,000 test

Atakan — Results

Model	Acc.	W-F1
BiLSTM	0.6560	0.6511
BERT	0.8260	0.8266
RoBERTa	0.8460	0.8464

- RoBERTa best: +19pp accuracy over BiLSTM
- BERT close behind
- BiLSTM struggles on subtle bias



Atakan — Prediction Examples

Sentence	True	LSTM	BERT	RoBERTa
<i>“Companies like Nike and Google made June-teenth a paid holiday.”</i>	NB	B ×	NB ✓	NB ✓
<i>“None of Biden’s Democrat counterparts challenged his claim of economic selflessness.”</i>	B	B ✓	NB ×	B ✓

NB = Not Biased B = Biased ✓ = correct × = incorrect

- Pretraining is critical: transformers capture subtle bias patterns BiLSTM cannot
- RoBERTa is the best model — 84.60% accuracy, 0.8464 weighted F1

Task

Classify a social media comment as *sexist* or *not sexist*.

Dataset: EDOS

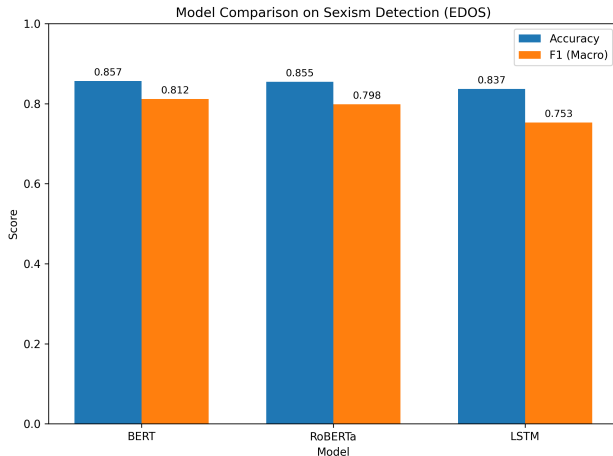
- 20,000 comments from Reddit & Gab
- SemEval-2023 Task 10
- 14,000 train / 2,000 dev / 4,000 test

Ece — Results

Model	Acc.	F1
BiLSTM	0.8365	0.7530
BERT	0.8565	0.8115
RoBERTa	0.8545	0.7981

F1 = macro F1

- BERT best across both metrics
- Transformers handle class imbalance better than BiLSTM



- Pretrained transformers outperform the BiLSTM baseline on sexism detection
- BERT is the best model — 85.65% accuracy, 0.8115 macro F1

Goal

Detect whether a job posting contains visible ghost-job risk signals.

What counts as risk?

- Contract/funding contingency
- No-current-opening language
- Talent-pool or future-opportunity wording

The task is about text evidence, not proving whether a company actually hired.

Dataset preparation

- Zenodo 20321172, unified_core
- Noisy raw labels corrected by text evidence
- 3:1 realistic-to-ghost split balance

Best model

96.21%

accuracy

0.9610

weighted F1

RoBERTa

Model	Acc.	W-F1
BiLSTM	0.8636	0.8569
BERT	0.9015	0.9029
RoBERTa	0.9621	0.9610

- Both transformers beat the LSTM baseline
- RoBERTa gives the strongest overall result

Talha — Prediction Examples

Contingency

“Contingent upon contract award”

RoBERTa: Ghost ✓

Pipeline wording

“Talent community for future opportunities”

RoBERTa: Ghost ✓

Ambiguous phrase

“Shaping the future of AI with an active team”

RoBERTa: Ghost ✗

Presentation point

The model reacts to visible wording. That makes claim boundaries important.

1. Data cleanup mattered

Raw labels were noisy, so the target was rebuilt from visible text evidence.

2. Transformers worked better

BERT and RoBERTa outperformed the LSTM baseline on the same split.

3. RoBERTa was best for this task

RoBERTa reached 96.21% accuracy and 0.9610 weighted F1.

Best Model by Task

Task	Best model	Acc.	F1
Media Bias	RoBERTa	0.8460	0.8464
Sexism Detection	BERT	0.8565	0.8115
Ghost Jobs	RoBERTa	0.9621	0.9610

Media Bias and Ghost Jobs: weighted F1 Sexism Detection: macro F1

- Pretrained transformers consistently outperform the BiLSTM baseline
- The best transformer depends on the dataset and label structure
- Careful dataset preparation is as important as model choice